

Digital Electronics Laboratory

Laboratory Exercise 2. The NOT, NAND and NOR Gates

Objectives:

- To investigate NOT, NAND and NOR gate operation.
- To study some the laws associated with the NOT, NAND and NOR operations.
- To investigate De Morgan's Theorem

Introduction

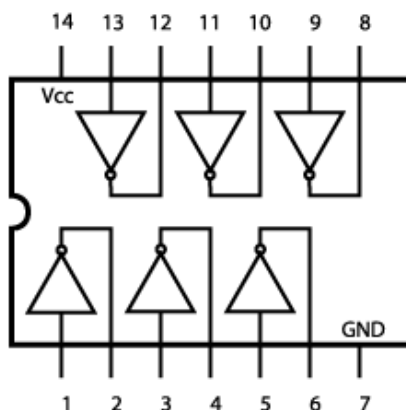
This is the second laboratory session, to allow you to become familiar with some more gates and the rules associated with them.

Pre-Laboratory

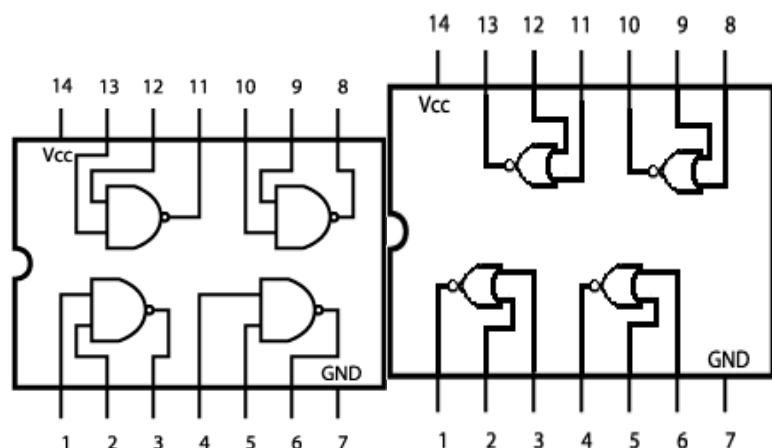
There are several tasks that you must perform **prior** to sitting this laboratory:

- Read the laboratory assignment in full.
- What is the relationship between the input and output of a NOT gate?
- Describe the NAND and NOR operation in your own words.
- Can any logic operation be performed using NAND gates? How?
- Can any logic operation be performed using NOT gates? How?
- State De Morgan's Theorem
- Draw up the truth table for a 2-input Exclusive-OR gate.

Useful Chip Diagrams:



7404(NOT)



7400(NAND)

7402(NOR)

The Laboratory:

Section 1. NOT Gate Implementation

(a) Connect one of the NOT gates (74LS04) as shown in Figure 1.

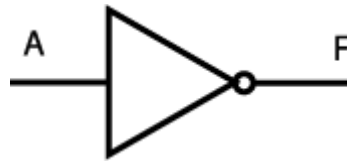


Figure 1. The NOT gate

(b) Vary the input to obtain all the possible combinations and complete the Truth table for the NOT gate.

(c) Connect two inverters to prove $\overline{\overline{A}} = A$. Describe how you did this, what did you record?

(d) Connect an AND gate and inverter as shown in Figure 2 to generate $A/\overline{A}$.

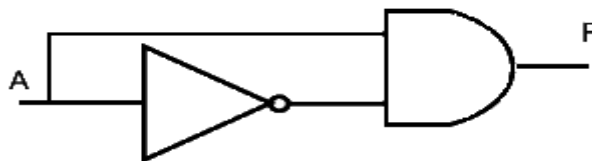


Figure 2. Circuit of equation $A/\overline{A}$

Change the states of A to note the effects. Describe what happens.

(e) Create a logic diagram to prove $\overline{A} + A = 1$. Implement this circuit and prove the law. Describe what happens in your circuit.

Section 2. De Morgan's Theorem

(a) Connect inputs A and B and their complements $\overline{A}$ and $\overline{B}$ to AND, OR gates as shown in Figure 3. You will have to use NOT gates to obtain these states.

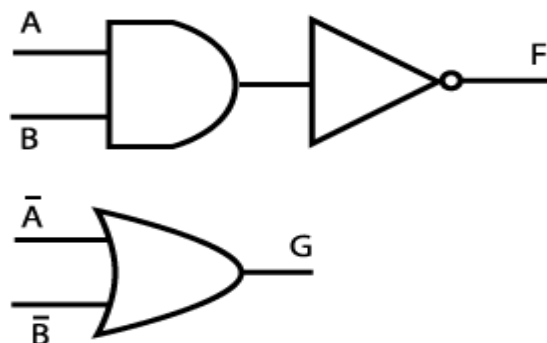


Figure 3. Proving De Morgan's Theorem.

(b) Vary inputs A and B and enter the output values into a truth table as show below. Also, what is the Boolean equation for F and G in terms of A and B?

A	B	F	G
0	0		
0	1		
1	0		
1	1		

(c) Use De Morgan's Theorem to minimise $F = \overline{A \cdot (\overline{A/B})}$, i.e.

$$F = \overline{A \cdot \overline{A \cdot B}}$$

What is the minimised equation. Implement the equation above and compare it to your minimised version.

Section 3. The implementation of the NAND gate.

(a) Connect one of the 2-input NAND gates as shown in Figure 4.

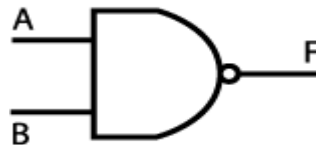


Figure 4. The NAND gate.

Enter the output values into a truth table as show below. Vary the inputs and record the output of the gate **F** . Examine the output of the NAND gate when one of the inputs is left floating (not connected). How can a NAND gate be used as an inverter?

A	B	F
0	0	
0	1	
1	0	
1	1	

Section 4. The implementation of the NOR gate.

(a) Connect the NOR gate as shown in Figure 5.

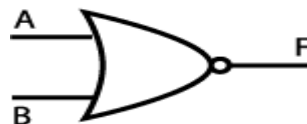


Figure 5. The NOR gate. Enter the output values into a truth table as show below. Vary the inputs and record the output of the gate **F** . Examine the output of the NOR gate when one of the inputs is left floating (not connected). How can a NOR gate be used as an inverter? Check your last answer on the breadboard.

A	B	F
0	0	
0	1	
1	0	
1	1	

Section 5. Conclusions

State briefly, but clearly, what you have gained from this laboratory.